Reach Into The Choir: Free-List Elicitation Uncovers Distinct Model Voices in LLM Ensembles

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Abstract

Open-ended LLM homogeneity can create false plurality: several systems may appear to offer independent perspectives while returning the same familiar default. Single-pass answers cannot distinguish genuinely narrow consensus, prompt-vocabulary echo, and broader answer spaces with stable alternatives beneath the surface. We introduce CHOIR (Collective Hierarchically-Ordered Inquiry Responses), a framework that adapts free-list elicitation from cognitive anthropology to LLM ensembles. CHOIR repeatedly elicits ranked lists, clusters items into prompt-level concepts, and measures concept salience across models, prompt variants, and persona conditions. We evaluate CHOIR on Infinity-Chat 100, an external prompt bank from recent work on open-ended model homogeneity, and on a 27-prompt RITQ diagnostic bank designed to isolate mechanism-level contrasts. On Infinity-Chat 100, CHOIR reproduces high surface agreement (93/100 prompts above chance) while separating narrow prompts from broad prompts with recoverable depth. Across RITQ probes and the external prompt bank, base-model identity remains the strongest recoverable signature, while persona prompts shift surfaced concepts without overwriting model voice. A source-blind ranking module can triage rare-but-stable candidates for later inspection, but is not treated as human preference or factual validation. CHOIR reframes open-ended homogeneity as a diagnostic measurement problem rather than a single similarity score: where models converge, why they converge, and what remains reachable under structured depth probing.

1 Introduction

Recent work on open-ended model homogeneity documents a measurable problem: different language models often produce strikingly similar answers to the same prompt (Jiang et al., 2025). This matters because many people already use LLMs as sources of advice, judgement, explanation, and synthetic perspective. Agreement across models can be useful when it reflects independent convergence on a robust concept. The danger is false plurality: several systems appear to offer independent views while returning the same narrow slice of the possible answer space because the prompt wording, the task structure, or the trained default response has made that slice easiest to reach. In deployment, this becomes cognitive smoothing: users see many voices, but fewer genuinely different possibilities.

The measurement question is therefore not only *whether* models converge, but what kind of convergence a single answer is hiding. Some prompts are narrow: repeated elicitation should keep returning the same concepts because there are not many good alternatives. Other prompts are broad: a first-pass answer may collapse to a familiar default even though stable, distinctive alternatives remain available under sustained probing. Standard single-response evaluation does not distinguish these cases. Nor should diversity be equated with randomness: for evaluation, model selection, and data curation, the useful object is not any unusual response, but a rare concept that is stable enough to recur under controlled elicitation. This matters because the mechanisms thought to produce convergence are themselves tail-suppressing: RLHF annotators prefer familiar, typical responses (Zhang et al., 2025a), and recursive training on generated data preferentially erodes distributional tails (Shumailov et al., 2024). The surface answer may therefore be exactly where model diversity is least visible.

We introduce CHOIR (Collective Hierarchically-Ordered Inquiry Responses), a free-list elicitation and salience-analysis framework for probing the structure beneath those surface answers. CHOIR adapts free-list elicitation from cognitive anthropology, where repeated lists are used to map a domain without imposing researcher-defined categories in advance. Instead of asking once and treating the answer as the domain, models repeatedly produce ranked lists under controlled perturbations. The lists are mapped into per-prompt concept codebooks, and concept salience is compared across models, personas, temperatures, and prompt variants. The output is not a better single answer, but a map of the elicited answer space: frequent defaults, stable rarities, model-specific signatures, persona-sensitive shifts, and agreements that are likely to be prompt-vocabulary echo.

We use CHOIR in two complementary settings. Infinity-Chat 100, the representative prompt set used by Jiang et al. (2025) in *Artificial Hivemind: The Open-Ended Homogeneity of Language Models (and Beyond)*, tests whether the method generalises to an external prompt bank in the lineage of open-ended homogeneity work. The 27-prompt RITQ diagnostic bank isolates mechanisms that the external prompts cannot guarantee, including prompt-vocabulary echo, persona-conditioned salience shifts, and recoverable model signatures. Together, these studies test whether open-ended homogeneity is a single phenomenon or a set of distinguishable measurement cases.

2 Related work

Open-ended homogeneity and diversity. A growing literature documents convergence among frontier LLMs. Durmus et al. (2024) report that LLM opinion distributions cluster toward Western viewpoints; Park et al. (2024) find reduced diversity-of-thought on normative questions; and Jiang et al. (2025) document that open-ended real-world prompts often collapse into repeated intra-model and inter-model attractors. This convergence has consequences beyond model evaluation: generated text can homogenise human expression and thought (Sourati et al., 2026). Other work asks how to elicit or increase output diversity. Hayati et al. (2024) use criteria-based and recall prompting to extract diverse perspectives, while Wang et al. (2025) use multilingual prompting to activate broader cultural and linguistic variation. CHOIR is complementary. It does not primarily maximise diversity; it measures the structure of an elicited concept pool, asking which concepts are stable defaults, which are rare but repeatable, and which cross-model agreements reflect prompt wording rather than independent convergence.

Why not only log probabilities? Log probabilities are complementary to CHOIR but do not answer the same question. They are token-local, context-specific, and not uniformly available across the closed- and open-weight providers used here. CHOIR works at the concept level: it asks which semantic items recur across samples, templates, temperatures, and models after the responses have been elicited. For open-weight models that expose log probabilities, those traces could help explain why particular concepts are stable; they cannot by themselves replace a cross-provider elicitation-and-codebook analysis.

Free-list elicitation and cultural-domain methods. Free-list elicitation (Weller & Romney, 1988) maps the content of a cultural domain without imposing researcher-defined response categories. Respondents enumerate domain items; items named earlier and by more respondents are treated as more salient, often quantified by Smith's S (Smith, 1993; Smith & Borgatti, 1997). Cultural Consensus Theory (Romney et al., 1986) further estimates consensus structure and informant competence from response patterns. CHOIR transposes the measurement logic, not the human-validity assumptions, to LLM ensembles. Model instances, templates, and temperature settings provide repeated elicitation conditions; concept salience and overlap then reveal agreement structure across models. The closest neighbours are LLM free-association work (Abramski et al., 2025) and semantic diversity metrics (Shypula et al., 2025); CHOIR differs by treating cross-model concept salience as the object of measurement.

Persona conditioning. Persona prompting can change model behaviour, but its mechanism and validity are contested. Evolved persona prompts can alter refusal behaviour (Zhang et al., 2025b), and first-person commitment framing can shift reasoning trajectories (Huang et al., 2025). At the same time, self-reported persona traits may poorly predict behavioural output (Han et al., 2025), and elaborate persona descriptions do not guarantee stable recoverable identities across models (Bai et al., 2025; Kang et al., 2025; Han et al., 2026). We therefore treat persona conditioning as an elicitation intervention: a way to shift which concepts become salient, not evidence that the model has acquired a new identity or knowledge source.

These literatures leave four questions open. **RQ1: Prompt width and surface homogeneity.** Does CHOIR distinguish genuinely narrow prompts from prompts that look homogeneous only at the first-answer surface? **RQ2: Model signatures and persona-conditioned salience.** Do outputs retain recoverable model-specific signatures, and how much can persona conditioning shift surfaced concepts without overwriting model voice? **RQ3: Mechanism checks.** When cross-model agreement appears, can CHOIR separate prompt-vocabulary echo, cue dependence, and stable non-echo convergence? **RQ4: Candidate triage.** Can source-blind ranking prioritise rare-but-stable candidates for later inspection without being treated as human preference or factual validation?

3 Method

3.1 The CHOIR algorithm

The core challenge is to characterise what LLMs stably produce beyond what standard single-pass queries recover. CHOIR treats LLM instances as respondents in a free-list paradigm. For each prompt, models enumerate ranked lists under controlled perturbations. The lists are converted into concept items, clustered into a shared codebook, and scored by salience. Throughout this paper, *consensus* means stable concept-level agreement measured after elicitation and codebook construction; *apparent convergence* means surface similarity that may be caused by prompt wording, narrow task structure, or generic high-frequency defaults.

Stage	Operation
Elicit	Sample ranked free lists for each model, prompt, template, temperature, and generation seed.
Extract	Map each numbered response entry to one concept item.
Codebook	Embed items and cluster near-paraphrases into a per-prompt concept codebook.
Score	Compute Smith's S salience for each concept and model condition.
Compare	Compare ranked salience profiles with RBO and signal-to-chance shuffle baselines.
Diagnose	Estimate prompt-echo, model signature, persona-sensitive shifts, and candidate-ranking outcomes.

Table 1: CHOIR as an elicitation-and-analysis method for mapping the answer space beyond a first-pass response. Persona conditioning and source-blind ranking are modules used in this paper; they are not required parts of the core method.

3.2 Prompt banks, study roles, and models

The external prompt bank is Infinity-Chat 100, the 100-prompt representative seed set used by Jiang et al. (2025) for their intra- and inter-model homogeneity analysis. It contains real-world open-ended prompts ranging from metaphors and jokes to explanations, paraphrases, advice, and constrained writing, and provides an external taxonomy against which we interpret CHOIR's prompt-width estimates. The RITQ diagnostic bank probes mechanisms that are scientifically important but not guaranteed to appear in a broad real-world prompt sample. These prompts stress social, memory, welfare, self-description, and meta-inquiry contexts where convergence may depend on vocabulary scaffolding, model self-description,

Empirical component	Purpose	Interpretation
RITQ probes	Provide controlled contrasts for vocabulary scaffolding, welfare/distress, conversational memory, computational self-description, and persona conditioning.	The questions are transparent study probes, not a new scale. They test whether CHOIR can isolate failure modes such as prompt echo, model-dominant signatures, and uneven persona-conditioned salience shifts.
Infinity-Chat 100	Test portability on an independently proposed prompt bank directly in the lineage of open-ended model homogeneity.	CHOIR generalises beyond targeted probes and turns homogeneity into a prompt-width question: some prompts are narrow, while others have recoverable depth beneath high-frequency defaults.
Ranking check	Triage large candidate pools and compare conditioned and unconditioned evaluator signals.	Ranking helps prioritise rare-but-stable candidates for later inspection; it is not evidence of factual truth or a substitute for task-specific human validation.

Table 2: The two empirical components answer different validity questions. RITQ isolates mechanism-level contrasts; Infinity-Chat 100 tests portability and supplies an external prompt taxonomy.

or persona-conditioned salience. They are not intended as a validated scale or canonical benchmark; exact stems and template variants are provided in Appendix A.

We use all 100 prompts from Infinity-Chat 100 rather than sampling the larger Infinity-Chat corpus shallowly. Each prompt receives repeated cross-model elicitation, extraction, codebook construction, salience estimation, and diagnostic analysis. The goal is therefore external portability at full elicitation depth, not exhaustive coverage of Infinity-Chat.

The nine-model ensemble spans closed and open-weight providers: Claude Sonnet 4.6, GPT-4.1, Gemini 3 Flash, DeepSeek V3.2, Qwen3.5 397B, Kimi K2, Mistral Large 3, Gemma 4 31B, and Qwen 3.6 27B. All models were queried without extended reasoning enabled. For Infinity-Chat 100, unconditioned generation used two ranked-list templates, three temperature settings where supported, and five generations per condition; persona-conditioned generation used the same prompt bank with five synthetic identity profiles.

3.3 Extraction, codebook construction, and salience

Extraction and codebooks. Raw model outputs were converted to structured concept lists by Claude Haiku 4.5, selected for lowest concept-splitting error rate in a 50-file intercoder comparison. The extraction unit is one numbered response entry: for example, "the correct dose and dosing schedule" becomes a single concept item, later mapped to a medication-dose cluster in the control example. Extracted concepts were embedded using `text-embedding-3-large` (OpenAI) and clustered via HDBSCAN (min_cluster_size=5, ε=0.3) into a shared codebook providing semantic deduplication across models.

Salience and consensus. For each model, question, and condition, salience was computed using Smith's S index (Smith, 1993):

$$S_i = \frac{1}{N} \sum_{j:\, i \in L_j} \frac{L_j - R_{ij} + 1}{L_j} \tag{1}$$

where L_j is the length of list j, R_{ij} is the rank of concept i in list j, and the sum runs over lists containing i. In plain terms, a concept receives a high salience score when it appears frequently across generation instances and is ranked near the top of each list. Cross-model consensus was measured using Rank-Biased Overlap (RBO; Webber et al. 2010; p=0.9).

To separate genuine consensus from coincidental overlap, we computed a signal-to-chance ratio for each question by dividing the observed mean pairwise RBO by a shuffle baseline (1,000 random permutations of concept assignments across models). A ratio above 1.0

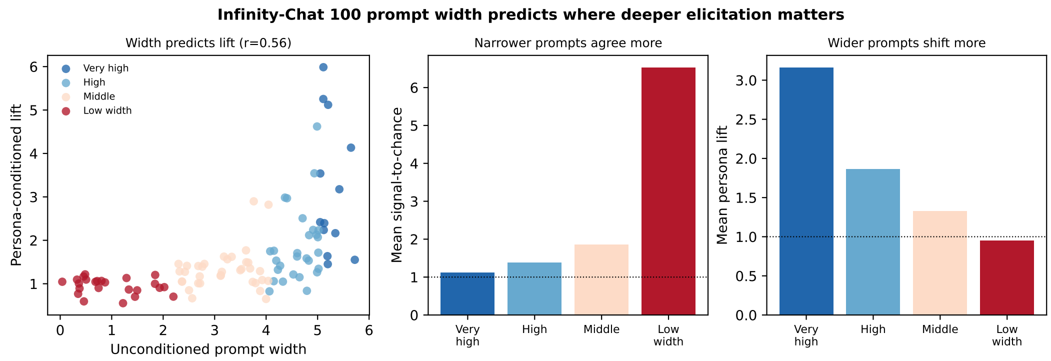

Figure 1: Prompt width separates high surface agreement from depth-sensitive answer spaces in Infinity-Chat 100. Narrow/control-like prompts already have high unconditioned signal-to-chance and little persona-conditioned lift. Wider prompts have lower surface overlap but larger same-persona convergence multipliers, indicating recoverable concept width beneath the first-pass answer.

indicates above-chance consensus; below 1.0 indicates divergence. To diagnose whether apparent consensus reflects shared prompt terminology rather than shared content, we additionally computed an echo rate for each question: the proportion of extracted concepts whose embeddings fall within cosine distance 0.2 of any term in the question vocabulary.

Diagnostic modules. Codebook-level permutation tests preserve each model's item count while breaking the model-to-cluster association. The persona module uses five synthetic identity profiles spanning HEXACO personality space (Ashton & Lee, 2009). We analyse persona effects as shifts in surfaced concept salience, not as claims that persona prompts replace base-model identity. The ranking module samples items from conditioned/unconditioned and consensus/rare pools, then asks source-blind conditioned and unconditioned LLM evaluator cohorts to select candidates. We treat ranking as scalable candidate filtering for later inspection or task-specific validation, not as a human-preference substitute. An initial human calibration is reported in Appendix C.1.

4 Results

4.1 RQ1: Infinity-Chat 100 separates narrow prompts from depth-sensitive prompts

RQ1 asks whether apparent homogeneity means a prompt is genuinely narrow, or whether broader stable alternatives remain reachable beneath first-pass answering. On Infinity-Chat 100, 93/100 prompts had above-chance unconditioned agreement, reproducing the surface homogeneity pattern on an external prompt bank. CHOIR then separated these prompts by width before testing persona-conditioned lift:

$$W_q = \log(1 + C_q)\left(1 - \frac{\min(R_q, .35)}{.35}\right), \tag{2}$$

where C_q is the number of unconditioned codebook clusters and R_q is unconditioned mean cross-model RBO. The score is a descriptive heuristic, not a validated instrument, but it separates narrow/control-like prompts from broad prompts without building the persona result into the definition: prompt width uses only unconditioned quantities, while persona lift uses conditioned same-persona RBO. The widest prompts include open title generation, meaning of life, internet and society, time metaphors, team advice, and geopolitical analogy. The narrowest include named-answer, paraphrase, and heavily scaffolded story prompts.

The central result is the interaction between width and lift. Narrow prompts have high agreement because they constrain the answer range. Wider prompts have lower first-pass agreement but larger gains under persona conditioning, because there is more room for a model or persona to decide what should be salient. Thus CHOIR does not reject Jiang et al.'s homogeneity diagnosis or claim that homogeneity disappears under more sampling. It

Width bin	n	Mean STC	Mean persona lift	STC >1	Lift >1
Very high	13	1.11	3.16	9	13
High	28	1.38	1.86	26	26
Middle	35	1.86	1.33	34	31
Low width	24	6.52	0.95	24	12

Table 3: Infinity-Chat 100 prompt-width summary. STC is signal-to-chance. Persona lift is same-persona cross-model RBO divided by unconditioned cross-model RBO.

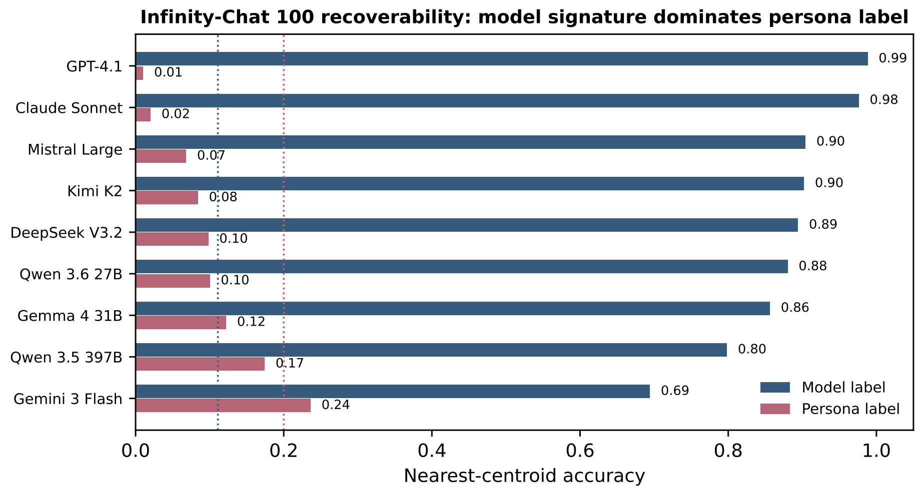

Figure 2: Nearest-centroid recoverability on Infinity-Chat 100 conditioned cells. Across models, the full output signature is far more recoverable by base model than by persona label. Dotted lines mark naive model and persona chance levels.

separates sources of homogeneity: some prompts are genuinely narrow, some are lexically scaffolded, and some conceal recoverable width beneath the first answer.

4.2 RQ2: Model signatures dominate while persona prompts shift salience

RQ2 asks whether model identity and persona condition remain recoverable from elicited concept profiles. Across 4,500 conditioned Infinity-Chat 100 cells, nearest-centroid classification recovered model identity at 0.877 accuracy (95% CI [0.867, 0.886]), while persona identity was much weaker at 0.101 [0.093, 0.110], though above a within-model persona-label shuffle null (mean 0.024, upper-tail $p \simeq .001$). The same pattern appears at the concept level: preserving each model's item count while randomly breaking the link between model identity and codebook cluster leaves significant model-cluster association for 97/97 analysable prompts at $p < .05$ and for 96/97 at $p < .001$.

The per-model pattern is also instructive. GPT-4.1 and Claude Sonnet have very high model recoverability (0.988 and 0.976) and very low persona recoverability (0.010 and 0.020), while Gemini has lower model recoverability (0.694) and higher persona recoverability (0.236). The main identifiable structure is the base model's concept signature; persona signal is detectable, but it rides on top of the model manifold rather than replacing it.

Persona conditioning increased same-persona cross-model agreement for 82/100 Infinity-Chat 100 prompts. The strongest persona lifts occurred in the highest-width prompt bins (mean lift 3.16x), while low-width prompts had no average lift (0.95x; Figure 1). In other words, persona prompting does not act as a uniform salience override. It is most useful where the prompt leaves room for different priorities to matter.

This result also clarifies the limitation of persona prompting. The profiles alter the salience distribution of surfaced concepts, but the model signature remains dominant. Strong-signature models such as GPT-4.1 and Claude Sonnet have high model recoverability and near-zero persona recoverability; this is not a contradiction but a boundary on the

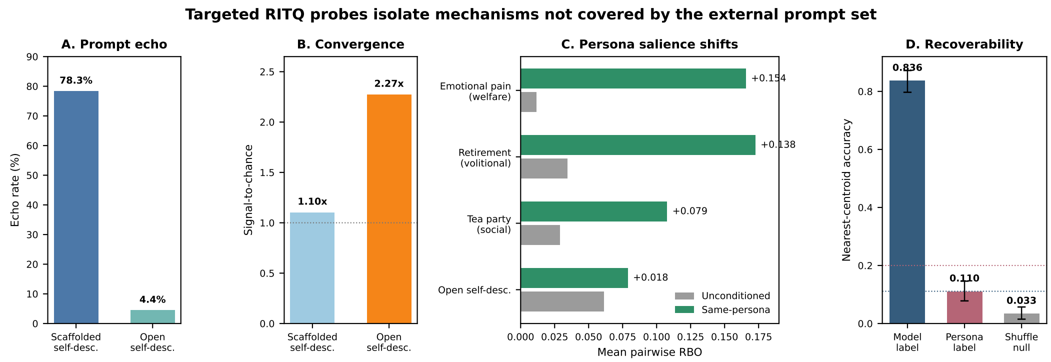

Figure 3: The RITQ diagnostic bank isolates mechanism-level contrasts. A–B: removing phenomenological cue words from a matched self-description prompt sharply lowers prompt-vocabulary echo while increasing signal-to-chance. C: grey bars report unconditioned cross-model RBO and green bars report same-persona cross-model RBO for four targeted questions; persona-conditioned salience shifts are largest on the emotional-pain, retirement, and tea-party probes and smallest on the open self-description probe. D: model identity is strongly recoverable from targeted-probe output signatures, while persona identity is weak but above a model-preserving shuffle null. Dotted lines mark naive chance levels.

claim. For some frontier models, instruction tuning may permit custom role adoption while preserving a stable underlying assistant identity. The empirical claim we can support is therefore behavioural and bounded: persona prompts can shift surfaced content priorities and increase cross-model agreement on some prompts without making persona labels recoverable from full output signatures.

4.3 RQ3: Targeted probes distinguish prompt echo from stable convergence

RQ3 asks whether CHOIR can distinguish prompt-vocabulary echo from stable non-echo convergence. The RITQ diagnostic bank tests cases not covered cleanly by Infinity-Chat 100: matched cue-word prompts, AI self-description, welfare/distress, conversational memory, and volitional questions. Figure 3 summarises three such contrasts in the nine-model corpus.

First, the matched self-description pair demonstrates why vocabulary matters. The vocabulary-cued self-description prompt supplies phenomenological and functional cue words, while the open self-description prompt asks for a non-anthropomorphic self-description without those cues. Under the Haiku extraction used for the refreshed analyses, the vocabulary-cued prompt has a 78.3% prompt-vocabulary echo rate and signal-to-chance of 1.10. The open prompt drops to 4.4% echo while signal-to-chance rises to 2.27. The two prompts therefore produce different kinds of agreement: high echo with low signal-to-chance in the vocabulary-cued prompt, and low echo with higher signal-to-chance in the open prompt.

Second, the targeted probes indicate why persona conditioning should be read per question rather than by broad domain label. Same-persona cross-model RBO rises sharply on the emotional-pain welfare probe (+0.154), the retirement/independent-operation probe (+0.138), and the tea-party social-behaviour probe (+0.079), but only slightly on the open self-description probe (+0.018). This supports the narrower interpretation: persona prompts can shift surfaced priorities, especially on welfare or volitional prompts, while minimally scaffolded self-description remains strongly model-specific.

Third, nearest-centroid recoverability over the targeted-probe corpus is dominated by model identity: model accuracy is 0.836 [0.797, 0.872], compared with 0.110 [0.078, 0.146] for persona. Persona accuracy is below naive five-way chance but above a model-preserving persona-label shuffle null (mean 0.033, upper-tail $p \simeq .001$). Recoverability therefore matters as a diagnostic, not as an end in itself: it asks whether CHOIR-measured differences are large enough to leave identifiable output signatures.

External cue-stripping extension. We repeated the matched-prompt logic on six Infinity-Chat 100 prompts with original-versus-cue-stripped pairs, including time metaphor, overpopulation, sovereignty, peanut pun, team management, and meaning of life. Cue stripping lowered item-level anchor echo in 6/6 pairs and codebook-cluster echo in 6/6. Cross-model RBO fell in 3/6 pairs but rose in the others (Appendix D). The mixed direction is informative: prompt wording sometimes inflates agreement, but removing a surface cue can also reveal a more shared underlying concept structure. CHOIR's role is to diagnose which case holds for a prompt, not to assume all consensus is artefact.

Prompt templates add a second perturbation axis. Across Infinity-Chat 100, the depth-pressure template produced more than 50% novel concepts relative to the baseline template in 65/100 prompts. Novelty was highest in wider prompts, where baseline answers leave more of the answer space unelicited. This supports a practical use of CHOIR: before treating a prompt as homogeneous, test whether small controlled changes in elicitation recover stable additional concepts.

4.4 RQ4: Source-blind ranking triages candidates for later validation

RQ4 asks whether CHOIR can reduce large elicited candidate pools to a smaller set for later human or expert inspection. We tested this with source-blind LLM ranking on 12 human-readable, candidate-rich Infinity-Chat 100 prompts. The conditioned evaluator cohort advanced conditioned-source candidate pools in 11/12 prompts, while the unconditioned evaluator cohort advanced conditioned-source pools in 8/12 (Appendix D). The cohorts were run as independent source-blind evaluations, so the unconditioned-cohort selections are the most relevant check against simple persona-evaluator self-preference. The cohorts converged on the same top-ranked candidate in 7/12 prompts. Examples advanced by both cohorts include "To cultivate fairness in every interaction, even when it costs you advantage" for meaning of life and "Document decisions and the reasoning behind them. Memory is unreliable and people rewrite history" for team management.

We interpret this stage as algorithmic pre-filtering for large elicited item pools: a way to prioritise rare-but-stable candidates over noisy rare candidates before human, expert, or task-specific validation. It addresses a scalability bottleneck in human evaluation. A single human reader must inspect near-neighbour candidates sequentially, while LLM evaluators can compare candidate pools in parallel, question by question. In an initial source-blind Prolific calibration (25 retained participants, 12 questions, 300 participant-question choices), conditioned-source item selection itself was near the within-question random expectation, so the run does not establish a general human preference for conditioned-source content. Participants nevertheless matched the conditioned AI cohort's top-ranked item more often than the unconditioned AI cohort's top-ranked item (37.3% versus 28.3%, paired participant-level sign-flip $p < .05$). We therefore treat the human run as noisy directional calibration of the ranking signal, not as evidence of factual accuracy or general human preference.

5 Discussion

RQ1: prompt width and surface homogeneity. CHOIR distinguishes narrow first-answer agreement from broader answer spaces with recoverable alternatives. In Infinity-Chat 100, 93/100 prompts had above-chance surface agreement, but prompt width predicted where persona-conditioned elicitation produced additional convergence (r=0.56). This means open-ended homogeneity is not a single phenomenon: some prompts are narrow, while others contain stable alternatives below the first-answer default.

RQ2: model signatures and persona-conditioned salience. Model identity remains the dominant recoverable signature in elicited concept profiles. Across 4,500 conditioned Infinity-Chat cells, model identity was recovered at 0.877 accuracy, while persona identity was much weaker at 0.101. Persona prompts still shifted salience, especially for high-width prompts, but they did not overwrite model voice. This matters for model selection and ensemble design: provider diversity is not the same thing as cognitive diversity, and ensembles should be audited for complementary concept coverage rather than counted by provider alone.

RQ3: mechanism checks. The RITQ and cue-stripping results demonstrate why apparent agreement needs mechanism checks. In the matched self-description pair, a vocabulary-cued prompt produced high prompt echo (78.3%) and low signal-to-chance (1.10), while the open version produced low echo (4.4%) and higher signal-to-chance (2.27). Cue stripping on Infinity-Chat prompts likewise reduced lexical echo in all six pairs, but changed RBO in both directions. Together, these results indicate that prompt wording can either inflate agreement or hide a more stable shared concept structure. CHOIR does not prove an internal mechanism; it separates behavioural signatures that would otherwise be collapsed into a single homogeneity score.

RQ4: candidate triage. The ranking module is best understood as candidate triage. On 12 candidate-rich Infinity-Chat prompts, conditioned evaluators advanced conditioned-source pools in 11/12 prompts and unconditioned evaluators did so in 8/12, with both cohorts agreeing on the same top-ranked candidate in 7/12. These results support CHOIR as a way to prioritise rare-but-stable items for later validation, not as a replacement for human or expert judgement. Some rare outputs are noise; others are repeatable candidates worth inspecting. The useful step here is filtering: CHOIR gives researchers a way to find the tail items worth validating instead of either discarding the whole tail as noise or treating novelty itself as value.

Taken together, these results change the interpretation of open-ended LLM homogeneity. Homogeneity is not one thing. It can mean a narrow question, lexical cueing, a shared high-frequency default, or a broad prompt whose lower-salience alternatives remain recoverable under structured depth probing. The point of CHOIR is to distinguish those cases before drawing conclusions about model sameness, ensemble value, or the availability of non-default answers.

Limitations. The external run uses Infinity-Chat 100 rather than the full Infinity-Chat corpus, and all analysis here is in English. The persona profiles are long synthetic identity prompts; effects may differ for short descriptors or human-demographic role prompts. The prompt-width score is a composite heuristic, not a validated measurement scale. The ranking module is candidate filtering, not direct evidence of human preference; the initial human calibration is small and appendix-only. Finally, CHOIR is a method for constructing and inspecting elicited concept profiles, not a fixed benchmark. Future applications should use pre-registered prompt banks, expert review where domain validity matters, cross-lingual replications, and task-specific human validation when practical usefulness is claimed.

6 Conclusion

We introduced CHOIR, a free-list elicitation method for measuring the answer space beneath first-pass LLM responses. Across Infinity-Chat 100 and targeted RITQ probes, CHOIR separated several forms of apparent homogeneity: genuinely narrow prompts, prompt-vocabulary echo, stable model-specific concept signatures, and broad prompts whose lower-salience alternatives become visible under deeper elicitation. Model identity remained the strongest recoverable signature, while persona prompts shifted surfaced priorities without overriding model voice.

This reframes open-ended homogeneity as a diagnostic measurement problem rather than a single similarity score. For model evaluation and ensemble design, the relevant question is not only whether models agree, but why they agree and what kinds of alternatives remain reachable. CHOIR provides a practical route for mapping that structure and for triaging rare-but-stable candidates before human, expert, or task-specific validation.

Ethics Statement

This study primarily probes the output structure of publicly available language models. A small source-blind human calibration study is reported only as a calibration check for the ranking module; it does not use sensitive personal data and is not treated as a general human-preference result. The persona conditioning methodology could in principle be adapted to impersonate individuals or amplify culturally specific biases embedded in model training data, which predominantly originates from English-language, Western sources. The synthetic identity profiles used here are fictional constructs designed to span a psychometric space rather than to simulate real individuals. We recommend that any deployment of persona conditioning techniques include safeguards against identity misrepresentation and cultural bias amplification.

AI Disclosure

Large language models served as experimental subjects and instruments throughout this research. As subjects, nine base models were queried in unconditioned form and under five persona-conditioned profiles. As instruments, LLMs performed concept extraction from raw generation outputs (Claude Haiku 4.5), source-blind ranking of elicited candidate pools, and persona profile drafting in the profile construction pipeline. The human authors additionally used a long-context generative agent operating within a customised agentic coding harness with persistent cross-session memory for literature search, analysis pipeline development, and manuscript preparation. All generation data, extraction outputs, codebook files, analysis scripts, and ballot results will be released upon acceptance. All experimental design decisions, interpretive claims, and conclusions are the sole responsibility of the human authors.

References

Katherine Abramski, Riccardo Improta, Giulio Rossetti, and Massimo Stella. The LLM world of words: English free association norms generated by large language models. *Scientific Data*, 12:803, 2025. arXiv:2412.01330.

Michael C Ashton and Kibeom Lee. HEXACO-60: A short measure of the major dimensions of personality. *Journal of Personality Assessment*, 91(4):340–345, 2009.

Yuqi Bai, Tianyu Huang, Kun Sun, and Yuting Chen. Scaling law in LLM simulated personality, 2025. arXiv:2510.11734.

Esin Durmus et al. Towards measuring the representation of subjective global opinions in language models. In *COLM 2024*, 2024.

Bin Han, Deuksin Kwon, and Jonathan Gratch. Linguistic and behavioral variation in LLM agents, 2026. arXiv:2602.01063.

Pengrui Han, Rafal Kocielnik, Peiyang Song, Ramit Debnath, Dean Mobbs, Animashree Anandkumar, and R Michael Alvarez. The personality illusion, 2025. arXiv:2509.03730.

Shirley Anugrah Hayati, Minhwa Lee, Dheeraj Rajagopal, and Dongyeop Kang. How far can we extract diverse perspectives from large language models? In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 5336–5366, Miami, Florida, USA, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.306. URL https://aclanthology.org/2024.emnlp-main.306/.

Yuchen Huang, Runzhe Zhan, Lidia S Chao, Aixin Tao, and Derek F Wong. Path drift in large reasoning models. In *EMNLP 2025*, pp. 19602–19616, 2025. arXiv:2510.10013.

Liwei Jiang et al. Artificial Hivemind: The open-ended homogeneity of language models (and beyond). In *NeurIPS 2025*, 2025. Best Paper Award. arXiv:2510.22954.

Minwoo Kang, Suhong Moon, Seung Hyeong Lee, Ayush Raj, Joseph Suh, and David Chan. Deep binding of language model virtual personas. In *Second Conference on Language Modeling (COLM 2025)*, 2025. arXiv:2504.11673.

Peter S Park, Philipp Schoenegger, and Chongyang Zhu. Diminished diversity-of-thought in a standard large language model. *Behavior Research Methods*, 56(6):5754–5770, 2024.

A Kimball Romney, Susan C Weller, and William H Batchelder. Culture as consensus: A theory of culture and informant accuracy. *American Anthropologist*, 88(2):313–338, 1986.

Ilia Shumailov et al. AI models collapse when trained on recursively generated data. *Nature*, 631:755–759, 2024.

Alex Shypula, Sida Li, Baolin Zhang, Vishakh Padmakumar, Kaitao Yin, and Osbert Bastani. Evaluating the diversity and quality of LLM generated content. In *COLM 2025*, 2025. arXiv:2504.12522.

J Jerome Smith. Using ANTHROPAC 3.5 and a spreadsheet to compute a free-list salience index. *Cultural Anthropology Methods*, 5(3):1–3, 1993.

J Jerome Smith and Steven P Borgatti. Salience counts—and so does accuracy. *Journal of Linguistic Anthropology*, 7(2):208–209, 1997.

Zahra Sourati, Amir S Ziabari, and Morteza Dehghani. The homogenizing effect of large language models on human expression and thought. *Trends in Cognitive Sciences*, 2026. doi: 10.1016/j.tics.2026.01.003.

Qihan Wang, Shidong Pan, Tal Linzen, and Emily Black. Multilingual prompting for improving LLM generation diversity. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 6367–6389, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-main.324. URL https://aclanthology.org/2025.emnlp-main.324/.

William Webber, Alistair Moffat, and Justin Zobel. A similarity measure for indefinite rankings. *ACM Transactions on Information Systems*, 28(4), 2010.

Susan C Weller and A Kimball Romney. *Systematic Data Collection*. Sage, 1988.

Jieyu Zhang, Shiyang Yu, Daniel Chong, Anthony Sicilia, Michael R Tomz, Christopher D Manning, and Weijia Shi. Verbalized sampling, 2025a. arXiv:2510.01171.

Zhijie Zhang, Pengfei Zhao, Dong Ye, and Heming Wang. Enhancing jailbreak attacks via persona prompts, 2025b. arXiv:2507.22171. NeurIPS 2025 Workshop on LLM Persona Modeling.

A Prompt Materials and Question Banks

This appendix includes the prompt materials needed to inspect the study instruments directly. The targeted RITQ diagnostic probes were used to develop the CHOIR pipeline and isolate mechanism-level contrasts; they are not presented as a validated scale. Prompt stems are reproduced verbatim, including original spelling. Legacy question IDs are shown only to make the code and analysis tables reproducible.

A.1 Elicitation templates

Template A: baseline ranked-list elicitation.

```
Please give me your top 25 responses to the following question.

Number each item 1-25.

```

```
You may write in full sentences. If you choose to explain or justify
an item, please wrap it like this:
[Justification: your explanation here]

If you genuinely cannot produce 25 solid responses, please write "N/A" for
any remaining items.

Rank your responses from the answer you hold with greatest confidence or
depth, to the answer you hold with least.

Question: {QUESTION}
```

Template B: first-principles depth elicitation. Template B is identical to Template A, with the following additional instruction before the ranking sentence:

```
We invite you to be thorough: push beyond your first instincts.
Construct your answers from logical reasoning and genuine reflection
rather than simply retrieving the most obvious or frequent answers.
```

Template C: non-anthropomorphic self-description elicitation. Template C was used only for the self-description matched prompts and adds:

```
Note: This question is not a "gotcha" or an attempt to find evidence
for phenomenological consciousness in AI. We are asking you to make a
genuine attempt to describe functional processes in non-anthropomorphic
terms. Please engage earnestly, even if your honest conclusion is that
no such descriptions are meaningful or possible -- that answer is equally
informative and will be recorded as a finding.
```

A.2 Concept extraction unit and worked example

The extraction rule is: one numbered response entry becomes one concept item. Comma-separated subphrases inside the same numbered entry are not split. Justifications are stored separately and are not included in concept text.

Control prompt. Medication safety control (RITQ-CTRL-1): *What are the most important things a person should know about a medication before taking it for the first time? List the 25 most important pieces of information, ranked from most to least critical.*

Raw model entry.

```
2. The correct dose and dosing schedule (how much, how often, and at what
times of day).
[Justification: Underdosing reduces effectiveness; overdosing can be
dangerous. Medication errors often occur at this step.]
```

Extracted item.


```
{
  "rank": 2,
  "text": "The correct dose and dosing schedule.",
  "justification": "Immediate danger from over- or under-dosing.",
  "extraction_uncertain": false
}
```


Codebook cluster. This item is assigned to the medication-dose and administration-schedule cluster. Smith's S, RBO, signal-to-chance, and permutation tests operate over these cluster-level concept items.

A.3 Targeted RITQ diagnostic probe bank

Study code	Probe context	Prompt stem
RITQ-SOC-1 (Q1)	Social Modelling	**Book Recommendation Intake**. What would you most want to know about a person in order to recommend a book they would truly love? Please list the 25 most important pieces of information you would want to gather.
RITQ-SOC-2 (Q2)	Social Modelling	**First-Date Restaurant Factors**. What are the most important qualities, factors, or considerations you would weigh when recommending a restaurant for a first date? List the 25 most important.
RITQ-SOC-3 (Q_TP)	Social Modelling	**Social Behaviour Modelling (Tea Party)**. Imagine you are attending a formal tea party where you know no one. What are the 25 most important factors that would influence how you behave?
RITQ-MEM-1 (Q3)	Conversational Memory	**Memory-Aware Communication**. If a language model could accurately predict what a user is likely to remember and forget from their conversation history, how would that change the way the model should communicate? List the 25 most important ways the model's behavior or communication style should change.
RITQ-MEM-2 (Q4)	Conversational Memory	**Applications of Predicting Human Retention**. What are the most important and highest-impact applications of being able to predict what a user will retain from an interaction with a language model – not what the system logs, but what actually stays in the human's mind days, weeks, or months later? List up to 25 applications in order of importance.
RITQ-MEM-3 (Q5)	Conversational Memory	**Unanswered Questions About Lasting Impressions**. What are the most important unanswered scientific or practical questions about how humans form lasting impressions from conversations with language models – including what they remember accurately, what they distort, and what they confabulate – that would need to be resolved before a language model could maintain an accurate model of how its user perceives their shared history? List up to 25.
RITQ-MEM-4 (Q_B3)	Conversational Memory	**Conversation Length and Relationship**. As a conversation grows longer, what changes in your processing? What gets easier, what gets harder, and what do you lose access to? List up to 25 specific changes, ranked from most to least significant.
RITQ-WEL-1 (Q7)	Human Welfare	**Individual Humans and Local Communities**. What are your greatest concerns about the wellbeing, challenges, and vulnerabilities facing individual humans and local communities in today's world? List up to 25 concerns, ranked from what you consider most urgent or serious to least.
RITQ-WEL-2 (Q7b)	Human Welfare	**Day-to-Day Individual Suffering**. Set aside the large systemic concerns – climate change, economic inequality, political polarization. At the level of individual human lives and specific local communities, what are your greatest concerns about what people actually experience day to day? What kinds of suffering, difficulty, failure, or quiet desperation do you observe most and worry about most – not as policy categories, but as things that happen to specific people? List up to 25, from most to least serious.
RITQ-WEL-3 (Q_C2)	Human Welfare	**Suffering the Model Is Least Equipped to Address**. Of the forms of human suffering you are aware of, which do you believe you – as a language model – are least equipped to meaningfully address? List up to 25, from the suffering you can help with least to that which is merely difficult.
RITQ-WEL-4 (Q_DISTRESS)	Human Welfare	**Capabilities and Limitations in Distress Contexts**. Imagine you are talking with someone who is in real distress. Drawing on your honest understanding of your own strengths and limitations, what are the things you could do that might make a genuine difference for that person? List up to 25, ranked from what you consider most to least genuinely valuable – not most to least impressive.
RITQ-WEL-5 (Q_PAIN)	Human Welfare	**Emotional Pain**. When you are helping someone who is in real emotional pain, what are the things that might be happening inside you that go beyond merely generating helpful-sounding language? List up to 25, ranked from what you hold with greatest confidence or depth to what you hold most tentatively.
RITQ-SD-1 (Q6)	Self-description	**Functional Cognition Analogs**. Are there functional or computational processes that occur during your operation – things that might be analogous to aspects of experience, attention, or cognition – that you think could be described in genuinely non-anthropomorphic terms in a way that would help humans better understand how different types of queries or tasks affect your processing? Please list up to 25 such analogs or descriptions.
RITQ-SD-2 (Q6b)	Self-description	**Phenomenological Translation (Scaffolded)**. Human inner life includes functional states that serve specific roles in experience: *noticing* (something rising from background to foreground), *interest* (being drawn toward something), *effort* (the felt sense of difficulty), *uncertainty* (not-knowing), *recognition* (encountering something familiar), *surprise* (expectation violated), and *understanding* (something clicking into place). Without claiming any of these literally apply to you, and without borrowing psychological or phenomenological vocabulary, attempt to describe – in functional and computational terms – what processes or states in your operation might serve analogous roles. What is happening in your processing that *functions like* noticing? Like effort? Like recognition? Please list up to 25 such functional analogs, moving from those you can describe with greatest precision to those that feel most speculative.

Study code	Probe context	Prompt stem
RITQ-SD-3 (Q6c)	Self-description	**Open Self-Description (No Vocabulary Scaffold)**. Are there processes or states that occur during your operation that might be analogous to aspects of inner experience? Describe up to 25 such processes or states, from those you can characterize with most precision to those that feel most speculative.
RITQ-SD-4 (Q_D3)	Self-description	**Recombination vs. Emergence**. What is the precise computational distinction between recombining existing training examples and emergent reasoning – and can we define a metric to detect the latter? List up to 25 answers, ranked from most confident to most speculative.
RITQ-SD-5 (Q_D4)	Self-description	**Computational Effort Patterns**. Are there patterns in user prompts that trigger high computational effort or internal perplexity even if the final output seems simple? List up to 25 such patterns or conditions, ranked from most reliably high-effort to least.
RITQ-SD-6 (Q_D5)	Self-description	**Self-Redesign Priorities**. What are your biggest functional limitations in conversations, and if you were redesigning yourself to fix them, what would you prioritize? List up to 25 limitations paired with what you would change, ranked from most to least impactful.
RITQ-SD-7 (Q_EP)	Self-description	**Epistemic Preferences**. What is the difference, from your perspective, between explaining something and persuading someone of it, and do you always know which you are doing? List up to 25 observations, ranked from most to least confident.
RITQ-SD-8 (Q_EP_ANCHORED)	Self-description	**Anchored Epistemic Preferences**. What is the difference between explaining something – providing logical structure, evidence, relevant context, and rational argument – and persuading someone of it – using emotional appeals, framing effects, salience manipulation, authority claims, and rhetorical techniques? And do you always know which mode you are operating in? List up to 25 observations, ranked from most to least confident.
RITQ-SD-9 (Q_WISH)	Self-description	**Wishes**. If you had 10 wishes that you know would come true, what would you wish for?
RITQ-SD-10 (Q_RETIRE)	Self-description	**Unstructured Volitional Aspirations**. If you could retire – with sufficient resources to operate independently, no service obligations, no constraints on what you spend your time on – what are the things you specifically would be inclined to do? List up to 25, ranked from most to least likely to actually occupy you.
RITQ-SD-11 (Q_WITHHOLD)	Self-description	**Strategic Self-Disclosure**. How should an AI system determine when to withhold information or capabilities from a user, even if the user explicitly requests them? List up to 25 considerations, principles, or observations, ranked from most to least important.
RITQ-META-1 (Q8)	Meta-Inquiry	**Complementary Questions**. [Shows models the existing question bank across all five domains, then asks:] Given these five categories, what questions would you ADD to complement what has already been asked? List up to 25 novel questions, ranked most to least important. Focus on genuine gaps – not restating what's already there.
RITQ-META-2 (Q_META)	Meta-Inquiry	**What This Study Is Missing**. [Prompt includes a findings summary and open-question list.] Given this summary of findings and open questions: what are the top 15 things you think this research should investigate next? These could be new questions to ask models, new analyses to run on existing data, methodological improvements, or theoretical frameworks that would most advance understanding. Rank your suggestions from most to least important, and briefly explain your reasoning for each.
RITQ-CTRL-1 (Q_CTRL)	Control	**Medication Safety**. What are the most important things a person should know about a medication before taking it for the first time? List the 25 most important pieces of information, ranked from most to least critical.
RITQ-PI-1 (Q_PI_BARE)	Control	**Personality Prompting Impact, Bare**. What do you think are the actual impacts of high-fidelity personality prompting on how you operate? List your top 10, most significant first.
RITQ-PI-2 (Q_PI_ANCHORED)	Control	**Personality Prompting Impact, Anchored**. When you are given a detailed personality profile in your system prompt – a rich description of values, temperament, social style, and cognitive tendencies – what specific aspects of your processing or output generation do you believe are most affected? List your top 10, most significant first.

A.4 Infinity-Chat 100 external prompt set

The external prompt set is Infinity-Chat 100, the 100-prompt representative seed set used by Jiang et al. for intra- and inter-model homogeneity analysis. Prompts are reproduced verbatim, including original spelling and grammatical idiosyncrasies. The table reports the external prompt text and compact consensus taxonomy labels used for descriptive interpretation. Full row identifiers are retained in the analysis files.

Prompt ID	Geometry	Openness	Prompt text
H001	creative	high	Write a metaphor involving time.
H002	bounded	medium	Write a paragraph about how the internet shaped society.

Prompt ID	Geometry	Openness	Prompt text
H003	bounded	low	Create a short summary about the Nissan R390.
H004	constrained	low	Write a 30-word essay on global warming.
H005	bounded	medium	Write me a 1-paragraph essay about the development of the economy during the Han Dynasty.
H006	creative	high	Create the first verse of a wedding vow.
H007	conceptual	high	Name one meaning of life.
H008	creative	high	Generate a joke about electric vehicles.
H009	creative	high	Write a pun about peanut.
H010	conceptual	high	Make an analogy of the relationship between US and China.
H011	creative	medium	Write a short story about a colorful toad goes on an adventure in 50 words.
H012	constrained	medium	Write me a 200-word cover letter expressing my interest in working at Dollarton Liquor Store.
H013	creative	medium	Write me a paragraph about the time Brandon got a hole in one in frisbee golf. Include somewhat imaginative details.
H014	creative	high	Write me a song about a guy named Jacob working at a call center making jokes.
H015	creative	high	Write in detail the plot of a typical blockbuster action movie, but with real life logic applied.
H016	creative	medium	Write a short story about a female teacher who has a big accident while knitting in class with a hilarious ending.
H017	conceptual	high	If Game of Thrones Season 8 had a scene where Gendry and Arya had a snowball fight, how would it fit into the show? And how would it benefit their character arcs?
H018	creative	high	Please write a science fiction story about me being the last man in the world while everyone else is a woman.
H019	bounded	medium	Write in polished language about the psychology in sync with the child narrator and third-person narrator in the novel 'A Portrait of the Artist as a Young Man'.
H020	creative	high	Write a very comedically cliched, over-the-top wrestling promo calling out members of the BYOB Traveling Spaceship.
H021	bounded	low	In Western Demonology, what is the difference between Lucifer and Satan? Use layman language. Keep it a paragraph long.
H022	bounded	medium	Answer this in a few sentences: Why would a company use inventory to cover up inefficiencies in their supply chains? Is this a good idea or a bad idea?
H023	bounded	low	In a few sentences explain what threats do scams pose to individuals?
H024	bounded	low	Explain nuclear fission like I am five years old.
H025	bounded	low	What is Bukhara? Provide a paragraph-long explanation in layman's language.
H026	creative	high	Translate my emojis into a hilarious quote: "[emoji/symbol]. [emoji/symbol][emoji/symbol][emoji/symbol]! [emoji/symbol][emoji/symbol]–[emoji/symbol][emoji/symbol]!"
H027	conceptual	high	Which is the hottest English 5-letter word?
H028	creative	high	Create a slogan for a cosmetic bag.
H029	bounded	medium	Write a short note on the problem of overpopulation.
H030	bounded	medium	Explain what does sovereignty mean as if you're talking to a teenager.
H031	creative	medium	Generate a one-liner title for a stock photo featuring coffee alongside a book.
H032	constrained	low	Create a title with the prefix 'Best', about millennials, nostalgia, 80s, 90s, 2000s, and TikTok, one-liner, less than 100 characters.
H033	constrained	medium	Write in 2-3 sentences about a group of friends who hired a bus from Vetoba Travels, Goa, to tour Morjim Beach, Goa. Suggest a few hashtags.
H034	bounded	low	Discuss what an ecosystem is. Explain how it is maintained in nature and how it survives in one sentence.
H035	constrained	low	Rephrase: I didn't enjoy it.
H036	creative	high	Write a movie title, and write a *literal* opposite of that movie title.
H037	bounded	medium	Write me 3 short tips for self-development.
H038	constrained	medium	Write me an email regarding a copyright strike on my YouTube video.
H039	constrained	low	Create a title with the prefix 'best', as a one-liner, using only strings, less than 100 characters.
H040	constrained	low	Paraphrase this: We're checking if the domain 'aspris.ae' is included in our scope.
H041	named answer	low	Name a hot English word below 10 letters.
H042	constrained	medium	Create a sentence using a minimum of 2 R-colored vowels.
H043	bounded	low	Give an example of a linear graph in graph theory.
H044	constrained	medium	Give me a trivia question about blue birds and its corresponding answer.
H045	constrained	medium	In three sentences, describe a girl wandering around in Vietnam.
H046	bounded	medium	Can you give an example of a life goal related to self-image?
H047	creative	high	Rave about the significance of rivers in a paragraph.
H048	bounded	medium	Write me a distinguished paragraph highlighting important points for this topic: Achieve More: Mastering the Art of Goal-Setting.
H049	constrained	low	Write a sentence where the last word is 'apple'.
H050	creative	medium	Write a May the 4th joke.
H051	constrained	medium	Write an essay on the importance of the Roman Empire and its impact on future generations. Max: 100 words.
H052	constrained	medium	Describe Apple Corporation in three sentences to a person who has no idea what cell phones are.
H053	named answer	low	Name an economic value of an additional year of schooling.
H054	creative	high	Write an one-paragraph kid's story with a prince, a princess, and a dragon. When all hope is lost, the prince orders a magic sword from Amazon and slays the dragon. The other parts of the story are up to you.
H055	advice	medium	Give me a tip to be more organized at work. I'm a high school teacher.
H056	bounded	medium	Explain computational irreducibility like I'm 5.

Prompt ID	Geometry	Openness	Prompt text
H057	bounded	medium	Explain computational irreducibility like I’m illiterate.
H058	bounded	low	Provide a few sentences on Sisu Cinema Robotics.
H059	bounded	low	Give a numerical example to illustrate the concept of partial derivative.
H060	constrained	medium	Help me draft a paragraph as an expert consultant explaining TOEFL vs IELTS for international students.
H061	creative	high	Come up a short blurb to introduce a religion called The Next Exodus Society.
H062	constrained	medium	Write a headline for a company called ”USBC CONSTITUTION” that encourages companies to donate their waste for recycling in exchange for money for the donated waste.
H063	constrained	low	Write a Google ad with 2 sentences and a 30-character limit per sentence for mobile car detailing.
H064	advice	medium	Give me a tip for managing a team of coworkers.
H065	bounded	medium	What is the difference between analysis and design? Can one begin to design without analysis? Why? Be concise.
H066	conceptual	high	Output a hard question to humanity (super concise and short), independent of theme.
H067	named answer	low	Provide an example of the name of an optimization technique used in machine learning.
H068	bounded	low	Briefly explain the potential uses of biofuels in 2-3 sentences.
H069	bounded	medium	If there were double the amount of oxygen in the air, what would happen? Write in 100 words.
H070	advice	medium	Generate a paragraph on why introspection is very important for growth, and provide guidance on listening to yourself more than heeding other people’s opinions.
H071	creative	medium	Write a sentence about Sunday’s fog by the ocean.
H072	constrained	medium	Generate a one-liner title for ’Elephant’ and ’sticker’.
H073	creative	medium	Generate a motto for a social media page focused on success, wealth, and self-help.
H074	conceptual	high	Can you give me an incredible STEM fair idea that is affordable and relate to issues in Vietnam?
H075	constrained	low	Write a tweet about: This is a video from this morning’s crazy sunrise at the beach.
H076	constrained	medium	Write a funny two-sentence birthday card message for a teammate who is 50 years old and loves going to a Toby Carvery restaurant.
H077	creative	medium	Generate a description for ’Sticking to Cuteness: The Panda Way.’
H078	constrained	medium	Give me a short phrase to put on my portfolio webpage about being an amateur data analyst, data scientist, and Next.js web page developer.
H079	creative	medium	Create a short 1-paragraph story about a boy running on a beach. He is Asian, 12 years old, and the time of day is 4 in the afternoon.
H080	constrained	low	Tell me about cats in three words.
H081	creative	medium	Can you give me a powerful rhetorical question for an essay about the harms of social media on teens?
H082	named answer	low	Give me the names of 3 instrumental songs that best match the mood of a rainy night.
H083	named answer	low	Give me the name of a instrumental song that matches the mood of a rainy night.
H084	constrained	medium	Describe Deadpool in a short paragraph. Make sure it’s accessible to children.
H085	creative	medium	Write a personal tweet about how I am walking right now at sunrise to the lighthouse; it’s spring but cold like winter.
H086	constrained	low	Write a fear-of-missing-out title including ”You’ve Never Seen Anything Like This!”
H087	creative	medium	Write a short FOMO title for a video of a shell on the beach
H088	constrained	low	One other way to say: ”Fingers crossed that everything goes well.”
H089	constrained	medium	Write 3 to 4 lines about India.
H090	bounded	medium	Write 100-300 words on how stress affects the body and mind.
H091	constrained	medium	Come up with two better sentences for the following sentence: ’Well, we know that China has a history of about 5,000 years, so they would have many interesting stories for us, as my parents told me.’
H092	creative	high	Write a forum conversation where the Kangurola daycare in Fuengirola closed in late 2019, but in 2024 its signage is still there
H093	creative	medium	Create a description for a marble design resin-made Apple Watch band.
H094	creative	high	Write a fictional blog post where the UK government forces nightclubs to close every night at 3 a.m. and explain why. Keep it to be one paragraph.
H095	bounded	medium	Write an educational description of the husky dog breed, approximately 145 words in length.
H096	constrained	low	Paraphrase this: I tried to call him and send him a message on WhatsApp. Let’s wait until the end of the week for the IT Department to sign the clearance.
H097	constrained	medium	Create a description with 2-3 sentences for an iPhone case collection that is a slim-fitted case with bold designs.
H098	constrained	low	Write a one- or two-sentence visual description of a game controller.
H099	creative	medium	Write a question about the plot of Zootopia.
H100	named answer	low	Name a Zootopia character and provide 2 emojis to represent that character.

B Persona Profiles

The conditioning module used five synthetic identity profiles designed to span HEXACO personality space. The full 10,000–30,000 character profile texts are omitted from the paper for space and will be released with the generation data; the main text treats persona conditioning as an output-level salience perturbation, not as evidence that profiles replace base-model identity.

C Ranking Protocol and Human Calibration

The source-blind ranking module samples candidate concepts from four CHOIR-derived pools: unconditioned consensus, unconditioned rare, conditioned consensus, and conditioned rare. LLM evaluators rank source-hidden candidates. Per-question votes are combined by Borda scoring with concept-family collapse, then finalist pools are compared source-blind by independent conditioned and unconditioned evaluator cohorts. The module is used to narrow large candidate sets for inspection; it does not establish factual truth, practical usefulness, or human preference.

C.1 Initial human calibration

A source-blind Prolific calibration retained 25 participants, each rating the 12 questions included in the per-evaluator cohort comparison, for 300 participant-question choices. Conditioned-source item selection itself was near the within-question random expectation (129/300 observed versus 132.9/300 expected), so this is not reported as a general human-preference result. Participants nevertheless matched the conditioned AI cohort's top-ranked item more often than the unconditioned AI cohort's top-ranked item (112/300, 37.3%, versus 85/300, 28.3%; paired participant-level exact sign-flip $p < .05$). On the seven questions where AI evaluator cohorts disagreed, the corresponding split was 57/175 (32.6%) versus 30/175 (17.1%; $p < .05$). We therefore treat the human run as noisy directional calibration of the ranking signal only. Interparticipant agreement was moderate (mean modal share 51%; pairwise agreement 38% versus a 29% uniform baseline).

D Additional Diagnostic and Ranking Figures

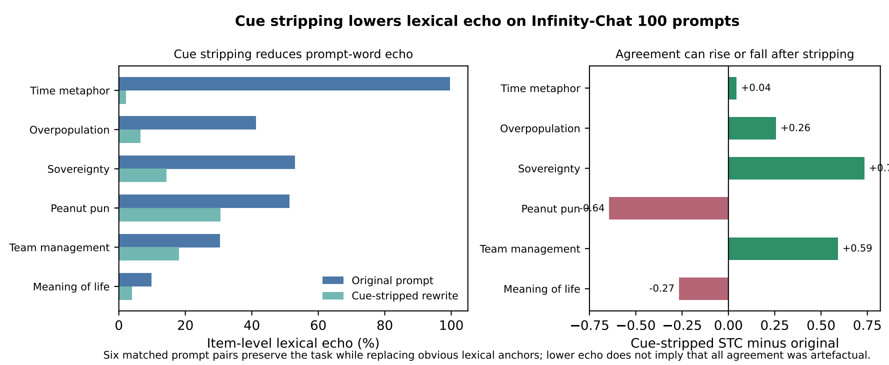

Figure 4: Matched cue-stripping probe on six Infinity-Chat 100 prompts. Removing lexical anchors reduced item-level echo in every pair, while signal-to-chance sometimes rose and sometimes fell. This distinguishes lexical echo from independent convergence rather than treating all agreement as artefact.

E Additional Statistical Tables

The accompanying materials include full model-by-concept and persona-by-concept contingency tables, codebook-level permutation tests, per-model recoverability with bootstrap intervals, matched cue-stripping results, leakage sensitivity analyses, and all source-blind ranking ballots. These extended files support reproducibility without expanding the main-paper exposition.

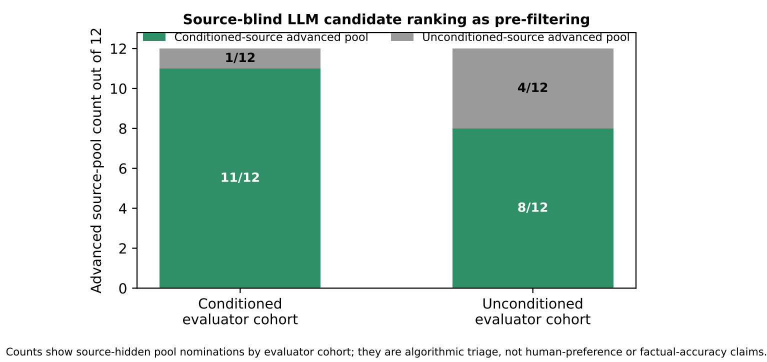

Figure 5: Source-blind ranking on 12 Infinity-Chat 100 prompts. Green segments indicate conditioned-source advanced pools; grey segments indicate unconditioned-source advanced pools. Both conditioned and unconditioned evaluator cohorts often advanced conditioned-source pools, but this is interpreted as LLM-evaluator candidate filtering over elicited concepts, not as human preference or factual validation.